Lezione 17 - 18/04/2024

Es 1

Calcolare la trasformata di Fourier dei seguenti segnali

a) **Sinc** s(t) = sinc(t)

b) **Rettangolo** scalato s(t) = rect(t/T)

$s(t) = \text{rect}\left(\frac{t}{T}\right)$

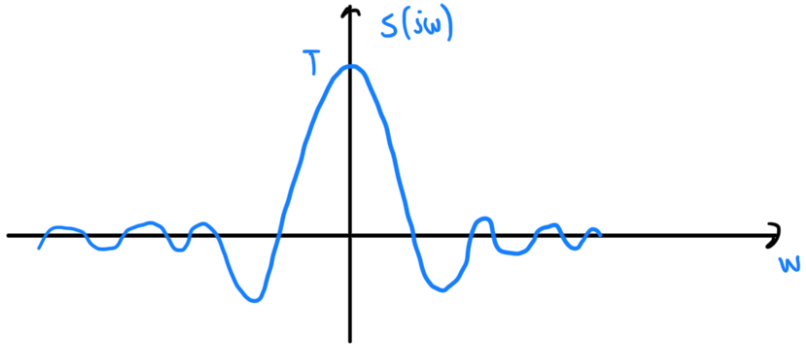
$S(j\omega) = ?$

SOL. APPLICHIAMO LA PROPRIETÀ DELLA SCALA

$$x(t) = \text{rect}(t) \xrightarrow{\mathcal{F}} X(j\omega) = \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$s(t) = x\left(\frac{t}{T}\right) \xrightarrow[\text{SCALA}]{\mathcal{F}} S(j\omega) = T\,x(jT\omega)$$

$$= T\,\text{sinc}\left(\frac{T\omega}{2\pi}\right) = T\,\text{sinc}\left(\frac{\omega}{2\pi/T}\right)$$



ESERCIZIO 1a

$s(t) = \text{sinc}(t)$

$S(j\omega) = ?$

SOL. APPLICHIAMO LA PROPRIETÀ DI SLIDE 54

$$\text{rect}(t) \xrightarrow{\mathcal{F}} \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$a = t$

$$\text{sinc}\left(\frac{t}{2\pi}\right) \xrightarrow{\mathcal{F}} 2\pi\,\text{rect}(-\omega) = 2\pi\,\text{rect}(\omega)$$

X CASA: provare per antitrasformata: $2\pi\,\text{rect}(\omega) \xrightarrow{\mathcal{F}^{-1}} \text{sinc}\left(\frac{t}{2\pi}\right)$

ABBIAMO SCOPERTO CHE:

$$x(t) = \text{sinc}\left(\frac{t}{2\pi}\right) \xrightarrow{\mathcal{F}} X(j\omega) = 2\pi\, \text{rect}(\omega)$$

$$s(t) = \text{sinc}(t) = x(t \cdot 2\pi) \xrightarrow{\mathcal{F}} S(j\omega) = a X(ja\omega)$$

SCALA CON $a = \frac{1}{2\pi}$

$$= \frac{1}{2\pi} 2\pi\, \text{rect}\left(\frac{\omega}{2\pi}\right)$$

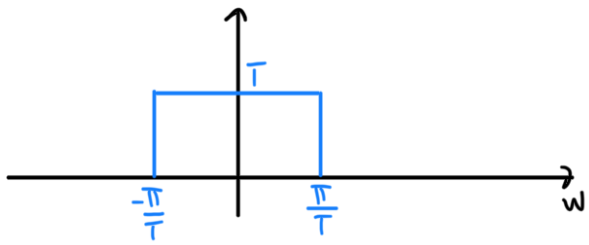
IN CONCLUSIONE ABBIAMO DIMOSTRATO CHE $\text{sinc}(t) \xrightarrow{\mathcal{F}} \text{rect}\left(\frac{\omega}{2\pi}\right)$

$$\text{rect}(t) \xrightarrow{\mathcal{F}} \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

sinc E rect SONO SEGNALI DUALI

ESERCIZIO 16 (ma con la scala)

$$\text{sinc}\left(\frac{t}{T}\right) \xrightarrow{\mathcal{F}} T\, \text{rect}\left(\frac{\omega T}{2\pi}\right)$$



[X CASA: $\text{sinc}\left(\frac{t-t_1}{T}\right) \xrightarrow{\mathcal{F}} ?$

SUGGERIMENTO: Uso regola di traslazione]

NOTA: SU ALCUNI ESERCIZI PROPOSTI SI USA LA NOTAZIONE IN F: LA TRASFORMATA È INDICATA COME $S(F)$

$$\omega = 2\pi F \longrightarrow S(j\omega) = S(F)\Big|_{F = \frac{\omega}{2\pi}}$$

$$F = \frac{\omega}{2\pi}$$

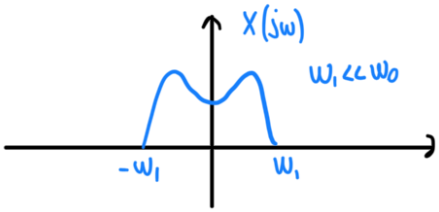
Es 1

Calcolare la trasformata di Fourier dei seguenti segnali

a) **Sinc** $s(t) = \mathrm{sinc}(t)$
b) **Rettangolo** scalato $s(t) = \mathrm{rect}(t/T)$
c) **Sinc** scalato $s(t) = \mathrm{sinc}(t/T)$
d) Segnale **costante** $s(t) = 1$
e) **Delta traslato** $s(t) = \delta(t-t_0)$
f) **Esponenziale** complesso $s(t) = e^{j\omega_0 t}$
g) **Sinusoide** $s(t) = \cos(\omega_0 t+\varphi_0)$ oppure $s(t) = \sin(\omega_0 t +\varphi_0)$
h) **Sinc quadro** $s(t) = \mathrm{sinc}^2(t/T)$
i) **Triangolo** $s(t) = \mathrm{triangle}(t/D)$
j) **Modulazione** double-side-band $s(t) = x(t) \cos(\omega_0 t)$
k) Convoluzione $s(t) = x*x(t)$ con $x(t) = e^{-at}\, 1(t)$, $a>0$
l) Convoluzione $s(t) = \mathrm{sinc}*\mathrm{sinc}(t)$
m) Trasformazione $s(t) = x(-2t+t_0)$
n) Segnale **segno** $s(t) = \mathrm{sign}(t)$
o) Segnale **gradino** $s(t) = 1(t)$

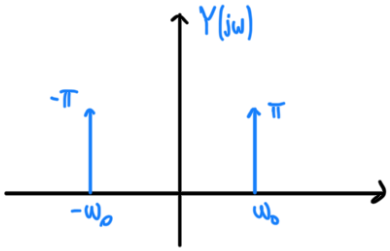
ESERCIZIO 1j (con le nuove regole di relazione tra prodotto e convoluzione)

$$s(t) = x(t)\underbrace{\cos(\omega_0 t)}_{y(t)}$$



$$x(t) \xrightarrow{\mathcal{F}} X(j\omega)$$

$$y(t) = \cos(\omega_0 t) \xrightarrow{\mathcal{F}} Y(j\omega) = \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$$

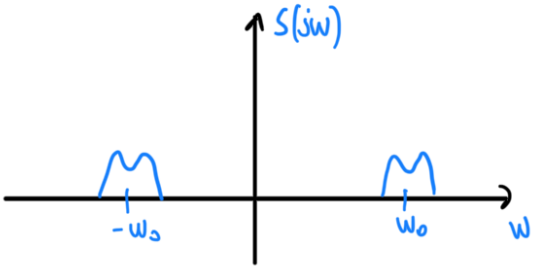


$$s(t) = x(t)y(t) \xrightarrow{\mathcal{F}} S(j\omega) = \frac{1}{2\pi} X * Y(j\omega)$$

$$S(j\omega) = \frac{1}{2\pi} X(j\omega) * \left[\pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)\right]$$

$$= \frac{1}{2\cancel{\pi}}\left[\cancel{\pi} X(j(\omega-\omega_0)) + \cancel{\pi} X(j(\omega+\omega_0))\right]$$

$$= \frac{1}{2} X(j(\omega-\omega_0)) + \frac{1}{2} X(j(\omega+\omega_0))$$



ESERCIZIO 1i

$$s(t) = \text{triangle}\left(\frac{t}{D}\right) \xrightarrow{\mathcal{F}} S(j\omega) = ?$$

SOL. MI RICORDO CHE IL TRIANGOLO È LA CONVOLUZIONE TRA 2 RECT

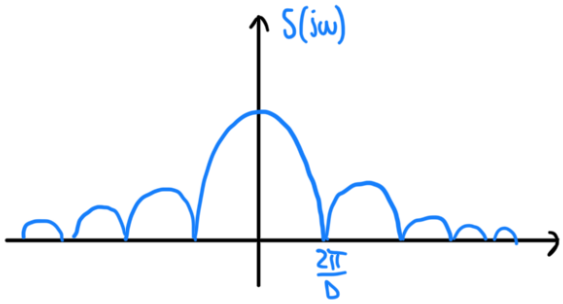
$$x(t) = \text{triangle}(t) = \text{rect} * \text{rect}(t) \xrightarrow{\mathcal{F}} \text{sinc}^2\left(\frac{\omega}{2\pi}\right)$$

REGOLA DI CONVOLUZIONE

$$\text{rect}(t) \xrightarrow{\mathcal{F}} \text{sinc}\left(\frac{\omega}{2\pi}\right)$$

$$s(t) = x\left(\frac{t}{D}\right) \xrightarrow{\mathcal{F}} S(j\omega) = D\,X(j\omega D) = D\,\text{sinc}^2\left(\frac{\omega D}{2\pi}\right)$$

SCALA $a = D$



ESERCIZIO

$$\text{sinc}^2(t) \xrightarrow{\mathcal{F}} ?$$

$$\text{triangle}(t) \xrightarrow{\mathcal{F}} \text{sinc}^2\left(\frac{\omega}{2\pi}\right)$$

$\omega = t$

$$\text{sinc}^2\left(\frac{t}{2\pi}\right) \xrightarrow{\mathcal{F}} 2\pi\ \text{triangle}(\omega)$$

$$x(t) = \text{sinc}^2\left(\frac{t}{2\pi}\right) \xrightarrow{\mathcal{F}} 2\pi\ \text{triangle}(\omega)$$

$$s(t) = \text{sinc}^2(t) = x(t \cdot 2\pi) \xrightarrow{\mathcal{F}} S(j\omega) = \frac{1}{2\pi} X\left(j\frac{\omega}{2\pi}\right) = \frac{1}{2\pi} 2\pi\ \text{triangle}\left(\frac{\omega}{2\pi}\right)$$

scala $a = \frac{1}{2\pi}$

$$\boxed{\begin{aligned} &\text{triangle}(t) \xrightarrow{\mathcal{F}} \text{sinc}^2\left(\frac{a}{2\pi}\right) \\ &\text{sinc}^2(t) \xrightarrow{\mathcal{F}} \text{triangle}\left(\frac{\omega}{2\pi}\right) \end{aligned}}$$

DUALITÀ TRIANGLE - SINC QUADRO

X CASA: $\text{sinc}^2\left(\frac{t-t_1}{T}\right) \xrightarrow{\mathcal{F}} ?$

$\text{triangle}\left(\frac{t-t_1}{T}\right) \xrightarrow{\mathcal{F}} ?$

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$s(t) = \text{sinc} * \text{sinc}(t)$

$S(j\omega) = ?$

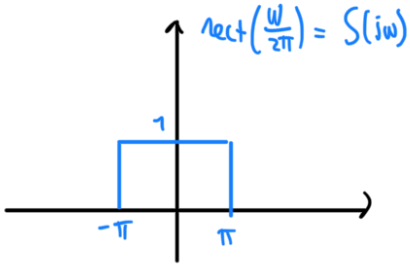
$s(t) = ?$

Sol. USO LA REGOLA CONVOLUZIONE - PRODOTTO

$$s(t) = \text{sinc} * \text{sinc}(t)$$

$$\mathcal{F}\downarrow \quad \mathcal{F}\downarrow \quad \mathcal{F}\downarrow$$

$$S(j\omega) = \text{rect}\left(\frac{\omega}{2\pi}\right) \cdot \text{rect}\left(\frac{\omega}{2\pi}\right)$$ REGOLA CONVOLUZIONE



IL RECT AL QUADRATO E' SEMPRE IL RECT (vale 1 tra $-\pi$ E π E 0 ALTROVE)

$$\text{sinc}(t) \xrightarrow{\mathcal{F}} \text{rect}\left(\frac{\omega}{2\pi}\right) = S(j\omega)$$

$$\mathcal{F}^{-1}$$

CIO' SIGNIFICA CHE $\text{sinc} * \text{sinc}(t) = \text{sinc}(t)$

X CASA: $z(t) = x * y(t)$ CON $x(t) = \text{sinc}\left(\frac{t}{2}\right)$
$y(t) = \text{sinc}\left(\frac{t}{3}\right)$

TROVARE: $Z(j\omega) = ?$
$z(t) = ?$ (SUGG: lo trovo per antitrasformata di $j\omega$)

Es 3

Calcolare l'area dei seguenti segnali

- sinc(t)

Es 4

Calcolare l'energia dei seguenti segnali

- sinc(t)

trovare A_s e E_s per $s(t) = \text{sinc}(t)$

Sol. APPLICO LE PROPRIETÀ DI AREA E ENERGIA (slide 57/58)

$$\text{sinc}(t) \xrightarrow{\mathcal{F}} \text{rect}\left(\frac{w}{2\pi}\right) = S(jw)$$

$$A_s = \int_{-\infty}^{+\infty} \text{sinc}(t)\,dt = S(jw)\Big|_{w=0} = \text{rect}\left(\frac{0}{2\pi}\right) = 1$$

$$E_s = \int_{-\infty}^{+\infty} \text{sinc}^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{+\infty} |S(jw)|^2\,dw = \frac{1}{2\pi}\int_{-\pi}^{\pi} \text{rect}^2\left(\frac{w}{2\pi}\right)dw = \frac{2\pi}{2\pi} = 1$$